

Helix: Accelerating Real-World Logistics

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Summary

Figure AI introduced a new application for their humanoid robots: logistics package manipulation and triaging. The post details improvements to Helix's System 1 (S1) visuo-motor control policy developed while iterating on this commercial use case.

Key Improvements

1. **Implicit stereo vision** – Helix S1 now has rich 3D understanding enabling more precise depth-aware motion.
2. **Multi-scale visual representation** – The policy captures fine-grained details while retaining scene level understanding.
3. **Learned visual proprioception** – Each robot can now calibrate itself, making cross-robot transfer seamless.
4. **Sport mode** – A test-time speedup technique allows Helix to achieve faster-than-demonstrator execution speed while maintaining high success rates.

The team also found that just 8 hours of well-curated demonstration data can yield a dexterous and flexible policy.

The Use Case

Package handling and sorting is a fundamental logistics operation involving transferring packages between conveyor belts while ensuring shipping labels are correctly oriented. Challenges include varying package sizes, shapes, weights, and rigidity. The system must determine optimal grasping moments for moving objects and handle environmental unpredictability with self-correction.

Architectural Improvements

Visual representation: The new S1 uses a stereo vision backbone with multiscale feature extraction. Features from both cameras are merged before tokenization, keeping computational

overhead constant.

Cross-robot transfer: A visual proprioception model estimates 6D end-effector poses from onboard visual input, enabling online self-calibration for strong cross-robot policy transfer.

Data curation: The team filtered demonstrations, excluding poor ones but keeping those with corrective behavior when failures were due to environmental stochasticity rather than operator error.

Sport mode mechanism: Action chunks representing T-millisecond trajectories are linearly resampled to shorter durations (e.g., $0.8 \times T$) and executed at the original 200 Hz control rate, yielding approximately 20% speedup without training modifications.

Results

Performance is measured using normalized effective throughput (T_{eff}), representing speed relative to demonstrator data.

- **Stereo impact:** The stereo model achieves a 60% increase in throughput over non-stereo baselines and generalizes to flat envelopes never seen in training.
- **Data quality:** A model trained with curated demonstrations achieves 40% better throughput despite being trained with 1/3 less data.
- **Sport mode:** Effective up to 50% speedup; beyond that, motions become imprecise. At 50% speedup, the policy achieves $T_{\text{eff}} > 1$, meaning faster than human demonstrators.
- **Cross-robot transfer:** The same policy trained on one robot's data maintained comparable performance across multiple additional robots.

Conclusion

The authors conclude that high-quality datasets combined with stereo multiscale vision, online calibration, and test-time speedup can achieve faster-than-demonstrator manipulation in real-world logistics using relatively modest amounts of demonstration data.